Sprint Planning

Sprint 1 – Project Setup

Sprint Goal

Establish the project foundation by defining the game concept, preparing the product backlog, and configuring the development environment and collaboration tools.

Sprint Length

2 weeks (February 21 – March 6)

Sprint Backlog

1. PB-01 Project Environment Setup
    - Create Unity project
    - Create GitHub repository
    - Verify Unity project compatibility across Mac and Windows systems
2. PB-02 Version Control Configuration
    - Research GitHub large file handling for Unity assets
    - Configure Git LFS for Unity asset management
3. PB-03 Game Concept Definition
    - Discuss and finalize initial game concept
    - Define core gameplay mechanics
4. PB-04 Backlog and Project Board Setup
    - Create initial product backlog
    - Write user stories
    - Create GitHub Kanban project board
    - Organize backlog items on the Kanban board
5. PB-05 Asset Research
    - Research and evaluate character sprite assets
    - Research and evaluate world/environment assets

Daily Scrum Notes

Meeting	Notes
1	Discussed game concept and core gameplay mechanics. Identified necessary development tools and workflow. Assigned responsibilities for backlog creation and asset research.
2	Configured Unity project and GitHub repository. Tested Unity project compatibility across Mac and Windows systems. Researched Git large file handling and configured Git LFS. Created backlog and GitHub Kanban board to track tasks.

Sprint Review

During Sprint 1 the team successfully established the technical and planning foundation for the project. The Unity project and GitHub repository were created and configured to support collaboration across Mac and Windows systems. Git large file handling was researched and configured to support Unity asset management. The team also defined the initial game concept, created the product backlog and user stories, and set up a GitHub Kanban board to manage development tasks. These activities prepared the project for gameplay feature development in future sprints.

Retrospectives

1. What went well
    - The development environment and version control were successfully configured.
    - The team established a clear game concept and gameplay direction.
    - The product backlog and Kanban board helped organize upcoming work.

2. What didn't go well
    - Environment setup and Git configuration took longer than expected.
    - Some time was spent researching tools rather than implementing gameplay features.

3. What can be improved
    - Plan technical setup tasks earlier in the sprint.
    - Begin implementing small gameplay features sooner.

4. Action Items / Next Steps
    - Begin implementing player movement mechanics.
    - Import selected assets into the Unity project.
    - Start development of core gameplay systems.

Sprint Progress

Sprint 1 focused on planning, environment setup, and project organization rather than gameplay implementation. The development environment, version control system, and collaboration tools were successfully configured. The product backlog and user stories were created, and asset research was completed. These preparations provide a strong foundation for implementing gameplay features in the next sprint. The updated Kanban board following sprint 1 can be viewed below.

Backlog 13 / 5

Estimate: 0

This item hasn't been started

- flap-and-fight #4
  Glide Control Option
- flap-and-fight #8
  Basic Attack Mechanic
- flap-and-fight #10
  Stationary Enemy Type
- flap-and-fight #15
  Game Over Screen
- flap-and-fight #19
  Power Ups
- flap-and-fight #18
  Entry Screen
- flap-and-fight #16
  Hit Feedback
- flap-and-fight #17
  Difficulty Scaling
- flap-and-fight #13
  Score System
- flap-and-fight #14
  Health Display HUD
- flap-and-fight #12
  Aggressive Dive Enemy
- flap-and-fight #11
  Patrol Enemy Type
- flap-and-fight #9
  Enemy Health System

Ready 6 Estimate: 0

This is ready to be picked up

- flap-and-fight #1
  Vertical Player Movement Control
- flap-and-fight #2
  Flap / Lift Mechanic
- flap-and-fight #3
  Gravity System
- flap-and-fight #5
  Horizontal Auto-Movement
- flap-and-fight #6
  Obstacle Spawning System
- flap-and-fight #7
  Collision Damage System

In progress 0 / 3

Estimate: 0

This is actively being worked on

In review 0 / 5

Estimate: 0

This item is in review

Done 5 Estimate: 0

This has been completed

- flap-and-fight #20
  Project Environment Setup
- flap-and-fight #21
  Version Control Configuration
- flap-and-fight #22
  Game Concept Definition
- flap-and-fight #23
  Backlog and Project Board Setup
- flap-and-fight #24
  Asset Research

Sprint 2 – Core Movement and Environment

Sprint Goal

Create the first playable prototype by implementing player movement mechanics, forward progression, and obstacle interaction.

Sprint Length

2 weeks (March 7 – March 20)

Sprint Backlog

1. PB-06 Vertical Player Movement Control
    - Implement vertical player movement control script
    - Test responsiveness of vertical player movement controls
2. PB-07 Flap / Lift Mechanic
    - Implement flap / lift input mechanic
    - Tune upward force values for flap mechanic
3. PB-08 Gravity System
    - Apply gravity physics to player character
    - Test gravity behaviour during gameplay
4. PB-10 Horizontal Auto-Movement
    - Implement automatic forward player movement
    - Test horizontal movement speed consistency
5. PB-11 Obstacle Spawning System
    - Create obstacle prefabs for environment hazards
    - Implement obstacle spawn timing system
    - Test obstacle spawning and spacing during gameplay
6. PB-12 Collision Damage System
    - Implement collision detection between player and obstacles
    - Apply damage to player when collisions occur

Daily Scrum Notes

Meeting	Notes
1	Discussed and agreed on player flap movement. Implemented vertical and horizontal movement. Added flap lift and needed animation to simulate a flapping bird.
2	Gravity system implemented. Play scene auto moves horizontally infinitely
3	The obstacles are now able to appear infinitely at random positions (from our generator). Collision with the pipes now results in immediate death.

Sprint Review

During Sprint 2 the team successfully implemented basic player movement with realistic physics. In addition to that, a good gravity system and flap mechanism was implemented so that players can move through the obstacles. As the player moves, the scene auto moves horizontally in parallel with the autogenerated pipes (obstacles). The obstacles are made so that when the player collides with them, it immediately dies regardless of the amount of hearts it had.

Retrospectives

1. What went well
   - Core gameplay mechanics were successfully implemented
   - A playable prototype was achieved by the end of the sprint

2. What didn't go well
   - Limited communication and collaboration during the sprint due to midterms
   - Obstacle spawning system is in place, but not yet optimal

3. What can be improved
   - Improve communication consistency between team members
   - Break tasks into smaller steps to better track progress

4. Action Items / Next Steps
   - Refine obstacle spawning system (timing and positioning)
   - Begin implementation of enemy and combat systems
   - Improve coordination and task distribution for next sprint

Sprint Progress

Sprint 2 focused on implementing the core gameplay systems required for a playable prototype. Most planned features were completed, including player movement, physics, and collision systems. The obstacle spawning system was partially implemented but requires further refinement. Progress during the sprint was impacted by midterm exams and reduced team availability; however, with midterms completed, improved productivity and communication are expected in the next sprint. The updated Kanban board following sprint 2 can be viewed below.

Backlog 6 / 5 Estimate: 0

This item hasn't been started

- flap-and-fight #4 — Glide Control Option
- flap-and-fight #15 — Game Over Screen
- flap-and-fight #19 — Power Ups
- flap-and-fight #18 — Entry Screen
- flap-and-fight #17 — Difficulty Scaling
- flap-and-fight #14 — Health Display HUD

Ready 7 Estimate: 0

This is ready to be picked up

- flap-and-fight #12 — Aggressive Dive Enemy
- flap-and-fight #16 — Hit Feedback
- flap-and-fight #11 — Patrol Enemy Type
- flap-and-fight #10 — Stationary Enemy Type
- flap-and-fight #9 — Enemy Health System
- flap-and-fight #13 — Score System
- flap-and-fight #8 — Basic Attack Mechanic

In progress 1 / 3 Estimate: 0

This is actively being worked on

- flap-and-fight #6 — Obstacle Spawning System

In review 0 / 5 Estimate: 0

This item is in review

Done 10 Estimate: 0

This has been completed

- flap-and-fight #20 — Project Environment Setup
- flap-and-fight #21 — Version Control Configuration
- flap-and-fight #22 — Game Concept Definition
- flap-and-fight #23 — Backlog and Project Board Setup
- flap-and-fight #24 — Asset Research
- flap-and-fight #2 — Flap / Lift Mechanic
- flap-and-fight #1 — Vertical Player Movement Control
- flap-and-fight #3 — Gravity System
- flap-and-fight #5 — Horizontal Auto-Movement
- flap-and-fight #7 — Collision Damage System

Sprint 3 – Combat and Enemy Systems

Sprint Goal

Introduce enemy interactions and combat mechanics to expand gameplay challenge.

Sprint Length

1 week (March 21 – March 27)

Sprint Backlog

1. PB-11 Obstacle Spawning System
    - Refine obstacle spawning rules for better user experience
    - Test obstacle spawning and spacing during gameplay
2. PB-13 Basic Attack Mechanic
    - Implement player attack input
    - Create attack animation or attack effect
3. PB-14 Enemy Health System
    - Implement enemy health tracking system
    - Apply damage to enemies when attacked
4. PB-15 Stationary Enemy Type
    - Create stationary enemy prefab
    - Implement stationary enemy collision behaviour
5. PB-16 Patrol Enemy Type
    - Implement patrol enemy movement pattern
    - Test patrol enemy interaction with player
6. PB-18 Score System
    - Implement score tracking system
    - Increase score when enemies are defeated
    - Increase score when obstacles are passed
7. PB-17 Aggressive Dive Enemy
    - Implement aggressive dive enemy behaviour
    - Test enemy dive attack behaviour
8. PB-21 Hit Feedback
    - Add visual or particle feedback when enemies are hit

Daily Scrum Notes

Meeting	Notes
1	Obstacles are now able to be generated randomly with constant spacing. Obstacles now move horizontally towards the player. Stationary enemy generation implementation Discuss on basic attack implementation. Decided on making it a bullet.
2	Basic attack implemented (bullets shooting from the player).

	Attack on enemy results in enemy to instantly die (for stationary ones) Implement a score system when a player passes in between the pipes. Discussion on non implemented items to return to the backlog. The score will be displayed up the top of the next to best score.

Sprint Review

During Sprint 3 the team successfully implemented obstacles for the game that the player has to avoid. The obstacles must be avoided at all cost just like the ground and the instant death for those components were implemented as well. In addition to that, stationary enemies are also randomly generated in the environment and are able to interact with the player. Attack methods were discussed, decided upon, and implemented. Attack on enemy and effect is implemented so that the enemy dies. Finally, a scoring system will be implemented and scores to be displayed with high scores.

Retrospectives

1. What went well
    - Obstacle generation and motion logic was implemented.
    - Clear communication on enemy attack was done with successful outcome and implementation.

2. What didn't go well
    - Due to the lack of time some of the items were not able to be implemented in the sprint and hence will be moved to the next sprint or will go back to the backlog to be implemented in a later sprint.

3. What can be improved
    - Consistent feature delivery throughout the sprint could improve so multiple features are not created all at once close to the sprint deadline.

4. Action Items / Next Steps
    - Move PB-16, PB-17 and PB-21 to sprint 5 since the next sprint is already planned and full and since developers will have capstone to prepare for.

Sprint Progress

Sprint 3 focused on implementing obstacle, attack and health related features. Probability and position of enemy appearance can be refined more. Communication was very effective this sprint and was an improvement from the last as developers worked hard to implement features. Most planned features were completed except 2 which will be transferred to the sprint after the next. The updated Kanban board following sprint 3 can be viewed below.

Backlog 0 / 5

Estimate: 0

This item hasn't been started

Ready 9 Estimate: 0

This is ready to be picked up

- flap-and-fight #12
  Aggressive Dive Enemy
- flap-and-fight #16
  Hit Feedback
- flap-and-fight #11
  Patrol Enemy Type
- flap-and-fight #14
  Health Display HUD
- flap-and-fight #15
  Game Over Screen
- flap-and-fight #17
  Difficulty Scaling
- flap-and-fight #18
  Entry Screen
- flap-and-fight #4
  Glide Control Option
- flap-and-fight #19
  Power Ups

In progress 0 / 3

Estimate: 0

This is actively being worked on

In review 0 / 5

Estimate: 0

This item is in review

Done 15 Estimate: 0

This has been completed

- flap-and-fight #20
  Project Environment Setup
- flap-and-fight #21
  Version Control Configuration
- flap-and-fight #22
  Game Concept Definition
- flap-and-fight #23
  Backlog and Project Board Setup
- flap-and-fight #24
  Asset Research
- flap-and-fight #2
  Flap / Lift Mechanic
- flap-and-fight #1
  Vertical Player Movement Control
- flap-and-fight #3
  Gravity System
- flap-and-fight #5
  Horizontal Auto-Movement
- flap-and-fight #7
  Collision Damage System
- flap-and-fight #6
  Obstacle Spawning System
- flap-and-fight #8
  Basic Attack Mechanic
- flap-and-fight #9
  Enemy Health System
- flap-and-fight #10
  Stationary Enemy Type
- flap-and-fight #13
  Score System

Sprint 4 – Gameplay Flow and UI

Sprint Goal

Complete gameplay systems and polish the player experience with scoring, user interface, and difficulty balancing.

Sprint Length

1 week (March 28 – April 3)

Sprint Backlog

1. PB-19 Health Display HUD
    a. Implement health display HUD
    b. Update health display when damage occurs
2. PB-20 Game Over Screen
    a. Implement game over condition when health reaches zero
    b. Create game over screen UI
3. PB-22 Difficulty Scaling
    a. Implement difficulty scaling system
    b. Adjust spawn rates or enemy difficulty over time
4. PB-23 Entry Screen
    a. Create entry/start screen for the game
    b. Implement start button to begin gameplay
5. PB-09 Glide Control Option
    a. Implement hold-to-glide mechanic
    b. Tune glide lift values for gameplay balance
6. PB-24 Power Ups
    a. Implement collectible power-up objects
    b. Apply temporary power-up effects to player

Daily Scrum Notes

Meeting	Notes
1	Game over screen implemented with sprite found from online free source. Hearts are implemented for the player to see how many lives they have. Difficulty scaling in terms of pipe and enemy generation and movement speeding up has been implemented. Powerups implemented with a base addition of 2 powerups (heart and shield). The heart powerup is tied to health and the scoring system. An entry screen with the player hovering has been implemented.

Sprint Review

During Sprint 4 the team successfully implemented features that were necessary to add more engagement to the game. Powerups were added making the game more playable and less difficult. Some display features such as entry screen, game over screen, and health display were added to aid players in the state of the game. The entry screen has the bird hovering so that the player has as much time they need before starting the game. The health display has 3 hearts past which the player will die. This feature ties to the heart collectible power up where it will be able to gain health if lost and if the heart state is full the heart will be converted to 1 point.

Retrospectives

1. What went well
    - The game is coming cohesively and with each feature added now we are adding more value to the game past base playable games.
    - Adding heart power up allows the player to play longer and as a result have fun.

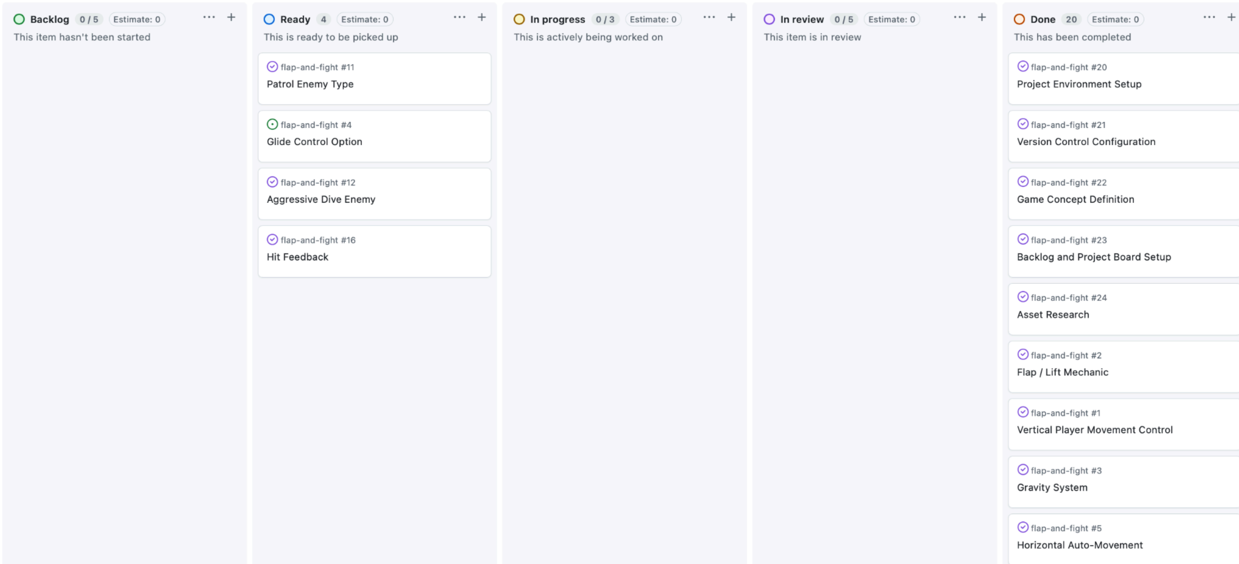
2. What didn't go well
    - There was nothing negative in terms of communication and feature implementations the team worked on.
    - One item was left un implemented because of time constraint passing it to the next sprint.

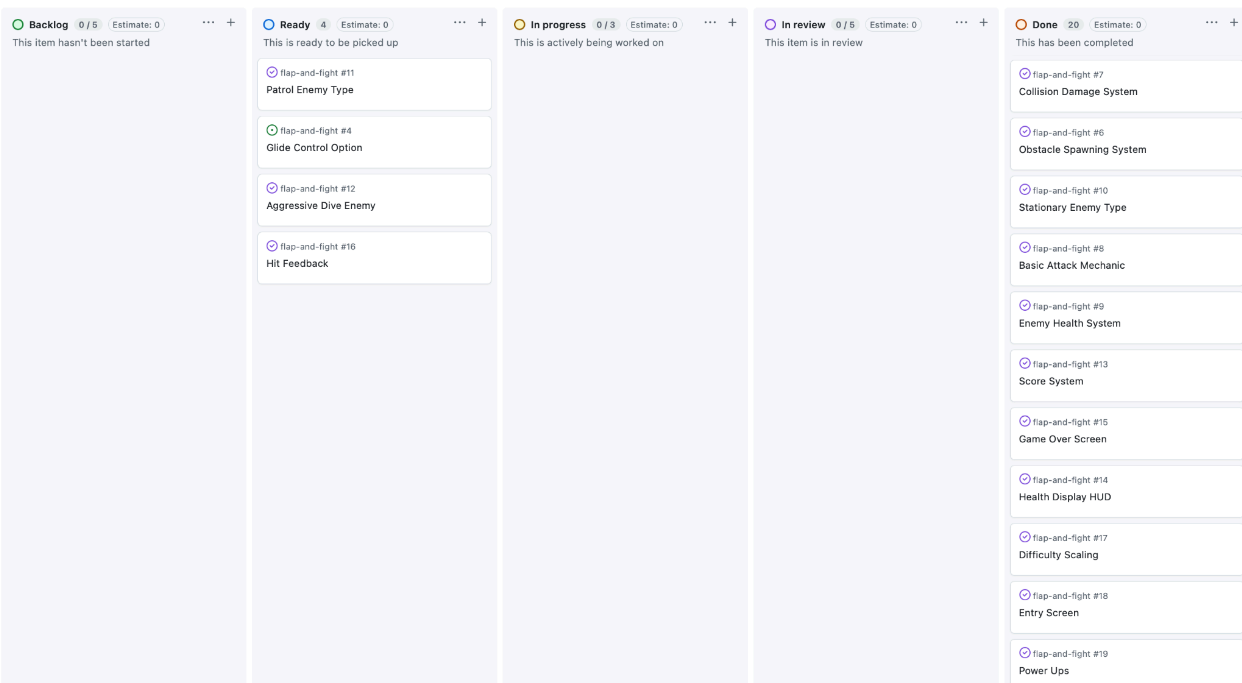
3. What can be improved
    - Incomplete items are observed more in the past sprints meaning the team should talk more before adding items and have detailed discussions regarding the possibility of completion.

4. Action Items / Next Steps
    - Pass the glide feature to the next sprint for implementation.
    - Get feedback from playtest results and player comments.

Sprint Progress

Sprint 4 focused on implementing mostly screen UI features to help players in having an easily navigable and informative experience. Most of the sprint backlog items were completed, all but 1 and that item will be moved to the next sprint where it will be implemented. The updated Kanban board following sprint 4 can be viewed below.


Backlog 0 / 5 Estimate: 0
This item hasn't been started
Ready 4 Estimate: 0
This is ready to be picked up
flap-and-fight #11
Patrol Enemy Type
flap-and-fight #4
Glide Control Option
flap-and-fight #12
Aggressive Dive Enemy
flap-and-fight #16
Hit Feedback
In progress 0 / 3 Estimate: 0
This is actively being worked on
In review 0 / 5 Estimate: 0
This item is in review
Done 20 Estimate: 0
This has been completed
flap-and-fight #20
Project Environment Setup
flap-and-fight #21
Version Control Configuration
flap-and-fight #22
Game Concept Definition
flap-and-fight #23
Backlog and Project Board Setup
flap-and-fight #24
Asset Research
flap-and-fight #2
Flap / Lift Mechanic
flap-and-fight #1
Vertical Player Movement Control
flap-and-fight #3
Gravity System
flap-and-fight #5
Horizontal Auto-Movement


Backlog 0 / 5 Estimate: 0
This item hasn't been started
Ready 4 Estimate: 0
This is ready to be picked up
flap-and-fight #11
Patrol Enemy Type
flap-and-fight #4
Glide Control Option
flap-and-fight #12
Aggressive Dive Enemy
flap-and-fight #16
Hit Feedback
In progress 0 / 3 Estimate: 0
This is actively being worked on
In review 0 / 5 Estimate: 0
This item is in review
Done 20 Estimate: 0
This has been completed
flap-and-fight #7
Collision Damage System
flap-and-fight #6
Obstacle Spawning System
flap-and-fight #10
Stationary Enemy Type
flap-and-fight #8
Basic Attack Mechanic
flap-and-fight #9
Enemy Health System
flap-and-fight #13
Score System
flap-and-fight #15
Game Over Screen
flap-and-fight #14
Health Display HUD
flap-and-fight #17
Difficulty Scaling
flap-and-fight #18
Entry Screen
flap-and-fight #19
Power Ups

Sprint 5 – Additional enhancing features

Sprint Goal

Add gameplay features that will add more

Sprint Length

6 days (April 4 – April 10)

Sprint Backlog

1. PB-09 Glide Control Option
   a. Implement hold-to-glide mechanic
   b. Tune glide lift values for gameplay balance
2. PB-16 Patrol Enemy Type
   a. Implement patrol enemy movement pattern
   b. Test patrol enemy interaction with player
3. PB-17 Aggressive Dive Enemy
   a. Implemented aggressive dive enemy behaviour
   b. Test enemy dive attack behaviour
4. PB-21 Hit Feedback
   a. Add visual or particle feedback when enemies are hit

Daily Scrum Notes

Meeting	Notes
1	When a player hits an enemy, there is a response feedback of an animation added when it dies. This is accompanied by sound as well. Aggressive moving enemy implemented.

Sprint Review

During Sprint 5 the team focused on polishing the combat experience and expanding enemy variety. A significant milestone was the addition of hit feedback, providing players with immediate visual and auditory confirmation of successful attacks through new death animations. Furthermore, the "Aggressive Dive Enemy" was fully implemented, introducing a new layer of difficulty that requires more active engagement than previous stationary types. However, the team decided that the "Patrol Enemy" was not as valuable, and decided to remove it from the design. These features, combined with the UI and power-up systems from Sprint 4, have brought the prototype closer to a final, integrated action game.

Retrospectives

1. What went well
   - Combat now feels more rewarding due to the implementation of hit feedback.

- The addition of moving enemies successfully addressed previous feedback regarding static gameplay.

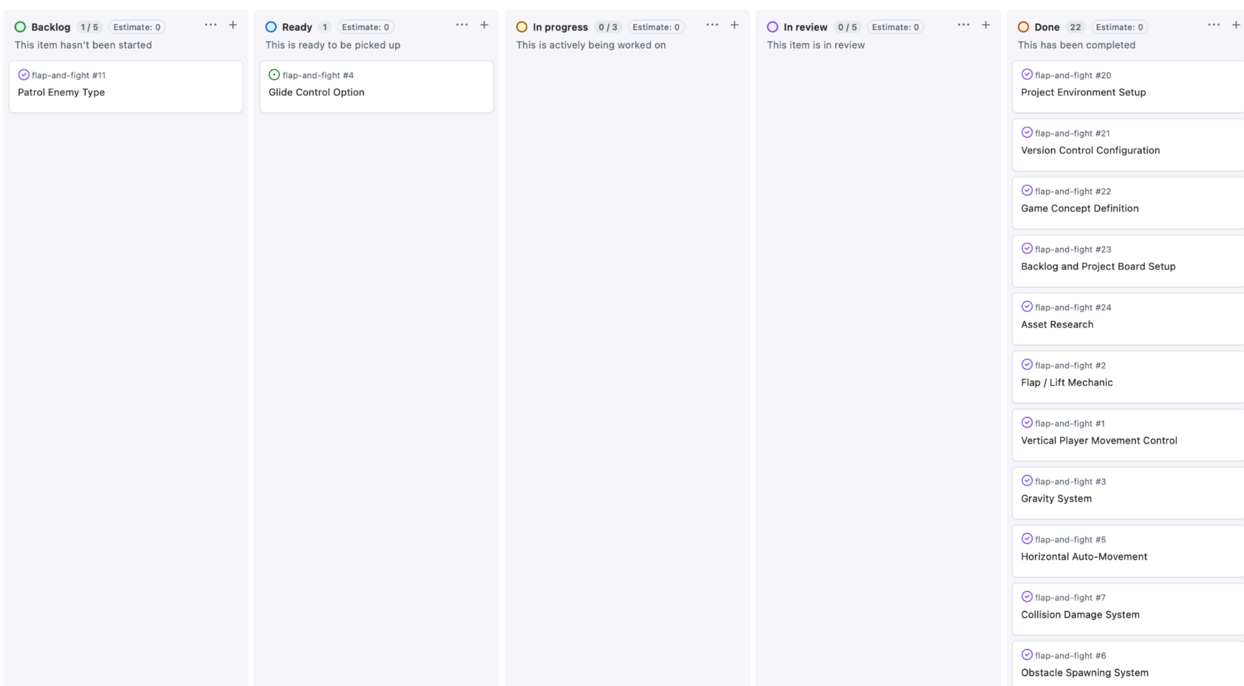
2. What didn't go well
   - The "Hold-to-Glide" mechanic (PB-09) proved technically complex to balance against existing gravity and remains a known limitation.

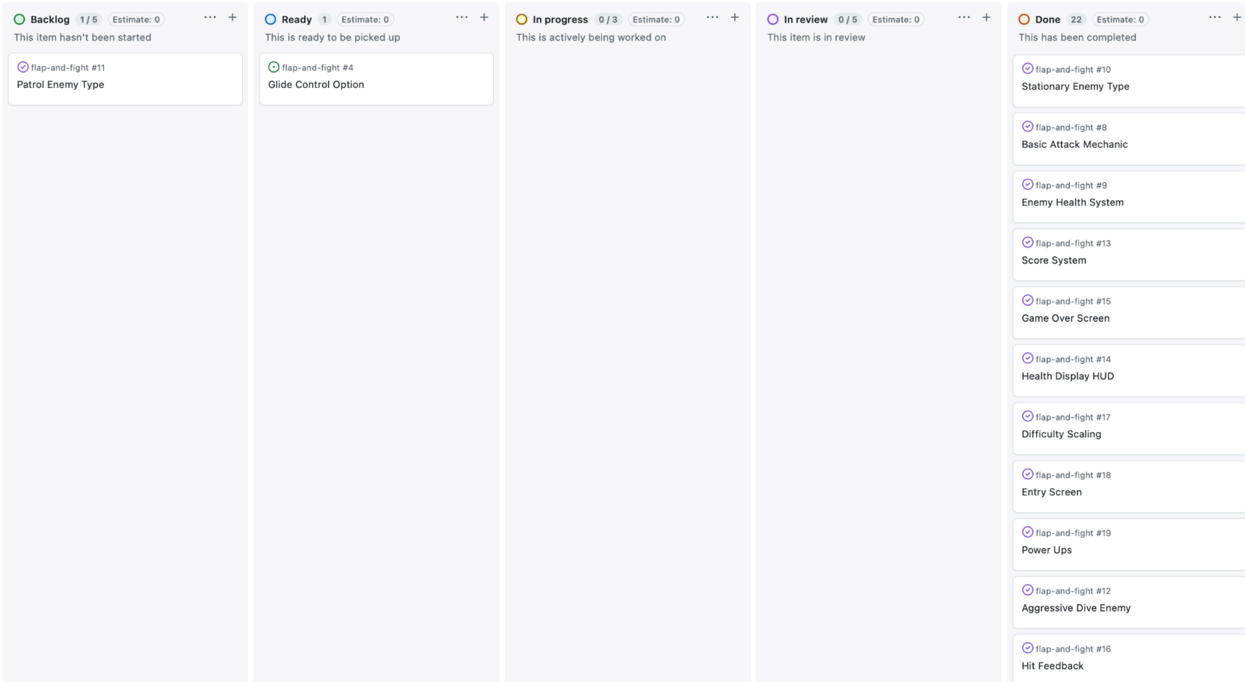
3. What can be improved
   - Better allocation of time for complex physics-based movement features to avoid carry-over into final deadlines.

4. Action Items / Next Steps
   - Finalize tuning for the glide mechanic.
   - Conduct a final review of the integrated systems based on playtest feedback.

Sprint Progress

Sprint 5 successfully addressed critical feedback regarding the lack of combat sensory feedback by adding death animations. The introduction of aggressive enemies has increased the mechanical depth of the game. While most enhancing features are now in place, the glide mechanic requires final implementation to ensure it does not negatively impact the established movement rhythm. The updated Kanban board following sprint 5 can be viewed below.




Backlog 1 / 5 Estimate: 0
This item hasn't been started
flap-and-fight #11
Patrol Enemy Type
Ready 1 Estimate: 0
This is ready to be picked up
flap-and-fight #4
Glide Control Option
In progress 0 / 3 Estimate: 0
This is actively being worked on
In review 0 / 5 Estimate: 0
This item is in review
Done 22 Estimate: 0
This has been completed
flap-and-fight #10
Stationary Enemy Type
flap-and-fight #8
Basic Attack Mechanic
flap-and-fight #9
Enemy Health System
flap-and-fight #13
Score System
flap-and-fight #15
Game Over Screen
flap-and-fight #14
Health Display HUD
flap-and-fight #17
Difficulty Scaling
flap-and-fight #18
Entry Screen
flap-and-fight #19
Power Ups
flap-and-fight #12
Aggressive Dive Enemy
flap-and-fight #16
Hit Feedback